Huawei Access Node Routing Protocol Flap Resolution Procedure for Whitefield East POP

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8755

Description

The failure domain of Routing Protocol Flap, specifically BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, on the Huawei access node WHI-ACC-50 at Whitefield East POP, is a critical issue that necessitates immediate attention to prevent significant network downtime and service disruption. This phenomenon occurs due to the complex interplay of routing protocols and their susceptibility to various factors such as network topology changes, configuration discrepancies, and equipment failures. In the context of Huawei equipment, the CLI dialect provides a robust set of commands to diagnose and resolve such issues. The underlying mechanics involve the continuous exchange of routing information between network devices, which, when interrupted or corrupted, can lead to routing instability. For instance, BGP route churn can occur due to inconsistent route advertisements, while OSPF adjacency loss might be caused by hello packet mismatches or interface failures. Understanding and addressing these protocol-specific issues requires a deep dive into the configuration, logs, and real-time protocol behavior using vendor-specific CLI commands.

Pre-requisite Checks

- Verify redundant link status to ensure no single point of failure
- Dispatch remote hands to Whitefield East POP for on-site assistance if needed
- Ensure all relevant network diagrams and documentation are up-to-date

Potential Root Causes

Cause	Probability
Inconsistent BGP route advertisements	High
OSPF hello packet mismatches	Medium
ISIS LSP flooding due to network topology changes	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp peer, Expected Output: List of BGP peers with their current status, Description: This command isolates issues in the BGP control plane by displaying the status of all BGP peers, helping to identify any peers that are not in the established state.

Command: display ospf interface, Expected Output: List of OSPF-enabled interfaces with their current state, Description: This command diagnoses OSPF adjacency issues by displaying the state of OSPF on each interface, including any errors or mismatches in hello packets.
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```
Command: display isis route, Expected Output: List of ISIS routes with their current status, Description: This command investigates ISIS LSP flooding by showing the current ISIS route table, allowing for the identification of any unstable or continuously updating routes.
```

Resolution Steps

Step 1: Step: Adjust BGP Route Advertisement, Command: bgp peer advertisement-interval , Description: Adjusting the BGP route advertisement interval can help stabilize BGP peers and reduce route churn. The specific interval value should be determined based on the network's requirements and stability considerations.

Step 2: Step: Correct OSPF Hello Packet Mismatches, Command: ospf area interface hello-interval , Description: Correcting hello packet mismatches involves adjusting the OSPF hello interval on the affected interfaces to match the network's design, thus stabilizing OSPF adjacencies.

Step 3: Step: Mitigate ISIS LSP Flooding, Command: isis timer lsp-gen-interval , Description: To mitigate ISIS LSP flooding, the LSP generation interval can be adjusted. This helps in reducing the frequency of LSP updates and thus stabilizes the ISIS protocol.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Stable BGP routing table, Description: Verifying the BGP routing table for stability after adjustments ensures that the BGP route churn issue has been resolved.

Command: display ospf neighbor, Expected Output: List of OSPF neighbors in the full state, Description: Checking the OSPF neighbor table verifies that OSPF adjacencies have been successfully established and are stable.

Command: display isis spf-log, Expected Output: ISIS SPF log showing no recent calculations, Description: The ISIS SPF log indicates the stability of the ISIS routing table. No recent SPF calculations suggest that the ISIS LSP flooding issue has been mitigated.
```

Roll-back Procedures

- Step: Revert BGP Configuration, Command: undo bgp peer advertisement-interval, Description: If the adjustment to the BGP route advertisement interval does not resolve the issue or introduces new problems, reverting this configuration change is necessary to return to the previous state.

- Step: Restore OSPF Interface Configuration, Command: undo ospf area interface hello-interval, Description: If adjusting the OSPF hello interval does not correct the adjacency issues or causes instability, restoring the original configuration is required to maintain network stability.

- Step: Reset ISIS Timer, Command: undo isis timer lsp-gen-interval, Description: If the adjustment to the ISIS LSP generation interval does not mitigate the LSP flooding or introduces unwanted side effects, resetting this timer to its default value is necessary for network stability.